Haardhik Ratnakar Kunder

01307, Dresden, Germany | kunderhaardhik@gmail.com | www.linkedin.com/in/haardhik-kunder/
github.com/HaardhikK | Portfolio: haardhik.netlify.app/

Education

TU Dresden, Dresden 2025 – Present
MSc in Computational Modeling and Simulation Applied AI track (3rd Semester)

University of Mumbai, Mumbai 2021 – 2025
Bachelor of Technology in Computer Engineering, Honors in Intelligent Computing

Projects and Research

Val Link Oct 2026
- Repurposed a **2019 phone with 4 GB RAM** into an offline AI companion running **three models locally** for speech recognition, thinking model and a vision model, supporting spoken conversations and questions about photos.
- **LoRA fine-tuned the thinking model** on app commands and connected the assistant to phone sensors and a downloaded **Dresden map and transit timetable**, enabling interactive responses, place search, and route planning without an internet connection.

Latent Communication for Multi-Trajectory Data Analysis Agents Blog Jun 2026
- Tested whether code-executing data-analysis agents can replace natural-language handoffs with shared hidden-state memory, comparing **single-agent**, **text based**, and **latent planner-coder-critic** modes.
- Showed latent coordination reduced decoded coordination tokens to **0** while text coordination grew from **979 to 1768 to 2677 tokens** across horizons, with short/medium latent pass rates of **1.00/1.00** versus text **0.93/0.87**.

Spending Rail TinyFish Accelerator Link Apr 2026
- Selected for **Phase 2 of the TinyFish Accelerator**, a 9-week program backed by Mango Capital with a **$2M seed funding pool**.
- Architected a **banking layer for autonomous agents** that provisions each agent with a scoped wallet and routes all purchases through a **single reconciliation layer**, eliminating manual spend oversight across multi-agent deployments.

Dresden-Live: 3D Geospatial Transit Engine [OpenSource] GitHub Mar 2026
- Architected a **3D live transit dashboard** using **Three.js** and **Mapbox GL JS** to visualize live VVO telemetry with **sub-second updates** across the Dresden transit network.
- Implemented resilient **async polling and buffering** to maintain interface responsiveness under API jitter, sustaining smooth rendering during high-frequency data bursts.

Publications

Exploring Deep Learning Techniques for Topic Modeling of Natural Disaster Tweets 2025
Springer Nature Link
- Analyzed tweet patterns during hurricanes, wildfires, and floods by engineering a custom **LLaMA3 + BERTopic** pipeline with **UMAP** and **HDBSCAN** clustering.

Experience

AI Engineering (Working Student), Snackz Aug 2026 – Present
Berlin, Germany (Remote)
- Develop agentic systems on **Gemini Flash**: a multi-step deep research agent (iterative web search, page reading, citation verification) delivering 20+ source-backed findings per task for under **$1.10 per run**, plus chained multi-agent workflows and an LLM-judged multimodal document pipeline.
- Build agent and tool orchestration for source discovery, information extraction and structured synthesis; used live A/B testing and cost telemetry to cut run costs by **20–70%** with no quality loss and speed up page processing **3**×.

Skills

Programming Languages: Python (Advanced), C/C++, JavaScript, TypeScript, MATLAB
Machine Learning: PyTorch, TensorFlow, scikit-learn, computer vision, statistical evaluation, data analysis
LLMs & Agents: LLM APIs, LangChain, Google ADK, RAG, FAISS, agent orchestration, LoRA fine-tuning
On-Device AI: Local inference, speech recognition, vision-language models, memory management, inference optimization
Software & Data: REST APIs, SQL, MySQL, PostgreSQL, Git, Docker, Linux, automated testing, debugging
Languages: English (C1), German (A2)